

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Pharm. Examination

University Theory Examination November 2017

PH217/PH229 Pharmaceutical Chemistry - III

Date: 27.11.2017, Monday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section I are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

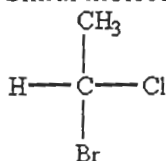
SECTION - I

Q 1 Attempt all questions. Each question is of one mark. 20

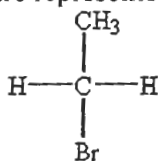
1. With respect to the electrophilic aromatic substitution of benzene which of the following is not true?
[A] A non-aromatic intermediate is formed
[B] Benzene acts as an electrophile
[C] A proton is lost in the final step
[D] Resonance forms are important
2. When considering electrophilic aromatic substitution reactions electron withdrawing substituents (e.g. nitro) are described as...
[A] *Ortho/para* directing and activating
[B] *Ortho/para* directing and deactivating
[C] *Meta* directing and activating
[D] *Meta* directing and deactivating
3. An aromatic molecule will have following characteristics. Except One
[A] Planar
[B] Cyclic Structure
[C] have $(4n+2)\pi$ electrons
[D] have $4n\pi$ electrons
4. Following are the facts about S_EAr Mechanism; EXCEPT ONE
[A] It is two step reaction
[B] Electron donating group on benzene ring increase the rate of reaction
[C] Electron withdrawing group on benzene ring increase the rate of reaction
[D] First step in mechanism is Rate determining step
5. Addition of Grignard reagent to aldehyde will gives _____
[A] 1° Alcohol
[B] 2° Alcohol
[C] 3° Alcohol
[D] All

6. Nucleophilic Acyl substitution reaction is fast with _____
 [A] Acid Chloride
 [B] Ether
 [C] Amine
 [D] Alkane

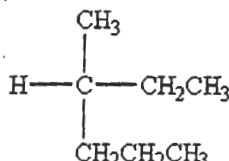
7. Chiral molecules are represented by:



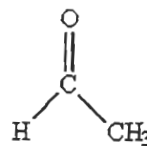
A



B



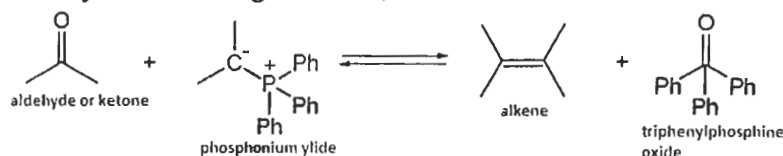
C



D

- [A] A, B, C and D
 [B] B, C and D only
 [C] A and C only
 [D] A, B and C only
8. Which of the following letter is recognized as Relative Configuration?
 [A] D and L
 [B] d and l
 [C] R and S
 [D] E and Z
9. Find out one of the correct statement from the options;
 [A] Aldehyde and Ketone gives substitution type of reaction
 [B] Nucleophilic substitution reaction is more faster at acyl carbon than saturated carbon
 [C] Ether is more reactive towards nucleophilic attack than ester
 [D] Amines is more reactive towards nucleophilic attack than amide
10. Which of the following is the definition of a meso compound?
 [A] A molecule with chirality centers which is not chiral
 [B] A molecule with chirality centers which is chiral
 [C] A chiral compound with more than one chirality center
 [D] A diastereomer with no chirality centers
11. In, Anthracene Electrophilic Aromatic Substitution Reaction Most Probably Occurs at ____ position
 [A] C-1
 [B] C-2
 [C] C-9
 [D] C-3
12. Following are the characteristics of the carboxylic acids. Except One
 [A] Dimer formation takes place
 [B] Electron withdrawing group increase the acidity of it
 [C] Polar in nature
 [D] Carboxylic acids have lower boiling point than alcohol
13. Which of the following statements is true?
 [A] All molecules that have chirality centers are chiral
 [B] All compounds with mirror images are enantiomers
 [C] Superimposable structures are enantiomers
 [D] Isomers that are not superimposable on their mirror images are enantiomers

14. In _____ phenolic compound intramolecular H- Bonding is Possible
 [A] p- Nitro Phenol
 [B] o- Nitro Phenol
 [C] p- Hydroxy Phenol
 [D] o- Bromo Phenol Chain
15. Cope rearrangement is an example of _____ type of reaction.
 [A] [3,3'] Sigmatropic Rearrangement
 [B] [1,3] Sigmatropic Rearrangement
 [C] [1,5] Sigmatropic Rearrangement
 [D] [5,5'] Sigmatropic Rearrangement
16. With Odd no of π Bond, The Electrocyclic Reaction under Thermal Condition in _____ fashion.
 [A] Conrotatory
 [B] Disrotatory
 [C] Antarafacial
 [D] Suprafacial
17. The term Atropiisomerism is used for.....
 [A] Biphenyl System
 [B] Spiran System
 [C] Heterocyclic System
 [D] Bridgehead bicyclic System
18. Anchimeric Assistance term associated with _____ mechanism.
 [A] S_N1
 [B] S_N2
 [C] S_EAr
 [D] Neighbouring Group
19. Identify the following reactions;



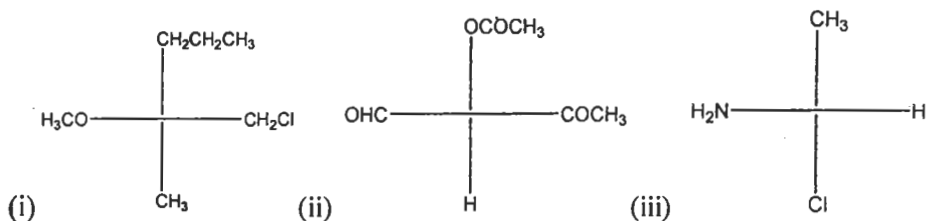
- [A] Wittig Reaction
 [B] Aldol Condensation
 [C] Reimer Tiemann Reaction
 [D] Cannizaro Reaction
20. Zinc- Amalgam is used in _____ type of reaction.
 [A] Wolff kishner Reduction
 [B] Aldol Condensation
 [C] Reimer Tiemann Reaction
 [D] Clemmensen Reduction

SECTION – II

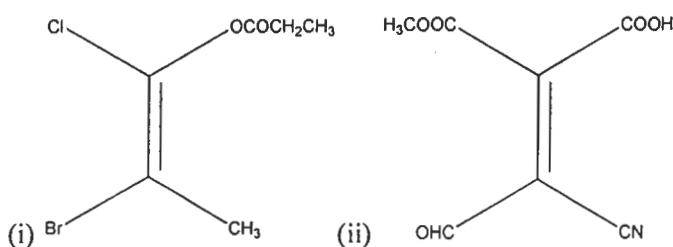
Q 2 Attempt any TWO of the following

- A Define Following Terms: (Any Five) 05**
- Optical Active Compound
 - Meso Compound
 - Cis and Trans
 - Racemic mixture
 - Relative Configuration
 - Erythro and Threo

B (I) Identify the R/S Configuration in following structures. (03 Marks) 05



(II) Identify the E/Z Configuration in following structures. (02 Marks)



C Write a note on Stereochemistry in Biphenyl System. 05

Q 3 Attempt any TWO of the following

A Compare Enantiomers Vs. Diastereomers 05

B Give a reason for following statement; 05

- Ester is more reactive than ether towards nucleophilic attack.
- Aldehyde and Ketones give addition type of reaction while carboxylic acid gives nucleophilic acyl substitution types of reaction.
- Trans form is more stable than cis form of any geometrical isomer.
- Cyclopentadiene is not aromatic but anion of it is aromatic.
- Staggered form of n butane is most stable than eclipsed form.

C Classify carboxylic acid derivatives with preparation of each derivative. 05

SECTION – III

Q 4 Attempt any FOUR of the following

A What is Conformation? Explain Conformation in n-Butane with energy diagramme. 05

B Write a note on stereoselective and stereospecific reaction. 05

C Write a short note on Neighbouring Group Mechanism. 05

D Write a Short Note on Electrocyclic Reaction. 05

E Write a Short Note on Sigmatropic Reaction. 05

F Short note on Orientation in Mono substituted Benzene 05

Q 5 Attempt any FOUR of the following

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|----------|---|-----------|
| A | Write Reaction and Mechanism of Fridal Craft Acylation Reaction of benzene. | 05 |
| B | Write any Three Preparations of Aldehyde and Ketone. | 05 |
| C | Explain Nitration of Benzene with suitable reaction and mechanism | 05 |
| D | Explain Aldol Condensation and Williamson's Synthesis with suitable reaction and mechanism. | 05 |
| E | Write a note on Cycloalkane. | 05 |
| F | Write Preparation and Reaction of Phenol. | 05 |
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